

Urban planning, the built environment, and subjective well-being: A review of recent research advances

Kostas Mouratidis

Department of Geosciences and Natural Resource Management, University of Copenhagen, Denmark

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ABSTRACT

Research on how the built environment shapes subjective well-being has expanded rapidly since 2021, providing new insights for urban planning and policy. Yet, no review has synthesized major recent advances in this field. This narrative review synthesizes five categories of advances – (1) disruptions to urban life, (2) mobility and accessibility, (3) social and experiential dimensions, (4) planning paradigms and urban form, and (5) urban artificial intelligence (AI) – based on a pathways framework linking the built environment to subjective well-being. Recent evidence highlights the importance of walkability, proximity, green space, and housing quality in supporting well-being, particularly under conditions of constrained mobility or when time spent at home increases. A composite of meeting places can foster social interaction and urban vitality, promote inclusion, and help combat loneliness. Public transport investments, including new metro systems and high-speed rail, are associated with social interaction, prevention of exclusion, and subjective well-being. Access to a private car may enhance activity participation and opportunities, particularly in car-oriented environments, while producing important equity and sustainability externalities. Architecture and public space design influence emotional well-being, with traditional styles linked to positive emotional responses. The 15-minute city, urban vitality, high-rise living, telework, and urban AI represent urban concepts and digital developments that transform urban planning and subjective well-being. The outcomes of the review strengthen and enrich the pathways between built environment and subjective well-being and raise attention to inequalities between population groups and contexts.

1. Introduction

With the rapid urbanization taking place worldwide, academic and professional interest in quality of life in cities is becoming a matter of global significance. Research on urban planning, the built environment, and subjective well-being (SWB) has been growing exponentially. Since the publication of a comprehensive review on the built environment and SWB in 2021 published in *Cities* (Mouratidis, 2021b), the field of urban well-being has seen a tremendous rise in scholarly output. There is more research on “urban” and “well-being” published from 2021 to the present than in all the years before 2021 combined (Fig. 1, left). Literature on the “built environment” and “well-being” has also more than doubled since 2021 (Fig. 1, right).

This rapidly expanding body of research has generated advances in the form of new empirical evidence on underexplored topics but also important conceptual and methodological advances in understanding the relationship between the built environment and SWB. The advances include: emerging topics like urban vitality and the 15-minute city, new

evidence from diverse locations including numerous studies from China, new evidence with more robust methodologies such as natural experiments and longitudinal studies, research on major disruptions to urban life such as the global pandemic and the rise of telework, new review studies on specific topics such as urban loneliness and car-based mobility, and new research themes due to the emergence of generative artificial intelligence that is transforming urban planning practice. These advances have expanded the field beyond establishing associations between the built environment and subjective well-being towards a more multidimensional understanding of the causal pathways between them.

Despite the rapid expansion of research on the built environment and SWB since 2021, there has been no review synthesizing the major conceptual, empirical, and methodological advances emerging during this period, and particularly within the context of pathways between the built environment and SWB. The only comprehensive recent review of pathways between the built environment and SWB focused on summarizing the influence of key built environment attributes without

E-mail address: km@ign.ku.dk.

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presenting recent advances in global research (Mouratidis, 2024a). In addition, there have been review studies focusing on particular population groups such as children (Kodali et al., 2023) and older adults (Mensi, Grotto, & Buja, 2026) or subtopics within built environment-SWB research such as green spaces and loneliness (Astell-Burt et al., 2022; Bower et al., 2023) or automobility and well-being (Miner, Smith, Jani, McNeill, & Gathorne-Hardy, 2024; Mouratidis, 2025a). There is therefore a substantial gap in knowledge pertaining to the advances in urban well-being literature since 2021.

This review paper attempts to cover this gap by investigating recent advances in literature on the built environment and SWB. The main aims of the paper are to: (1) identify key advances in research on the built environment and SWB since 2021, (2) synthesize new knowledge from these advances within an urban planning and policy perspective, and (3) suggest promising research avenues. The paper is based on a qualitative narrative review approach. The review is organized around the pathways linking the built environment to SWB, as conceptualized in Mouratidis (2021b). Findings from the reviewed studies are presented and discussed in relation to an updated version of this conceptual model.

The contribution of the paper goes beyond updating the literature. It synthesizes how recent research has expanded and changed our understanding of pathways between the built environment and SWB. In particular, the review integrates conceptual advances in the form of emerging themes, methodological advances, and new empirical evidence within a pathways framework relevant to urban planning and policy.

The paper is structured as follows. Section 2 presents the methodology followed in the paper. Section 3 presents the conceptual model and an overview of the research advances. Sections 4–8 review the identified five advances. Section 9 discusses the findings and provides concluding remarks.

2. Method

The paper adopts a qualitative narrative review approach to synthesize recent advances in research on the built environment and SWB. A narrative review was considered more suitable than a systematic review because the field is extremely broad and rapidly expanding. Moreover, the field includes diverse themes, conceptual approaches, and methodologies. The aim of the review is not to provide an exhaustive inventory of all studies published since 2021, but to identify and synthesize major conceptual, empirical, and methodological advances within the context of pathways linking the built environment and SWB. The approach is therefore interpretive and theory-informed, following the conceptual framework developed by Mouratidis (2021b). The use of the framework provides a theoretical basis for organizing and interpreting a broad and rapidly expanding literature, although, as with other narrative and theory-informed reviews, this approach may place greater emphasis on

studies and themes most relevant to the adopted framework. Similar qualitative narrative review approaches have been used in previous reviews of broad urban and well-being themes (Mouratidis, 2021a, 2021b, 2024c; Nordbakke & Schwanen, 2014).

The review primarily covers studies published between January 2021 and February 2026 in international peer-reviewed journals. The identified studies are discussed in relation to pre-2021 literature. Literature searches were conducted iteratively using Scopus and Google Scholar and were complemented by backward snowballing from relevant review papers and influential studies. Searches focused on combinations of keywords related to the built environment, urban planning, and subjective well-being, as well as specific subtopics identified during the review process. Indicative search terms used to identify subtopics included combinations such as “built environment” AND “well-being” and “urban” AND “well-being”. Additional targeted searches were conducted for subtopics that were identified through the review process. Indicative search terms included combinations such as “covid” AND “urban” AND “well-being”; “public transport” AND “well-being”; “car” AND “well-being”; “telework” AND “well-being”; “built environment” AND “telework”; “built environment” AND “social interaction”; “built environment” AND “emotional experience”; “density” AND “well-being”; “compact city” AND “well-being”; “urban-rural” AND “well-being”; “spatial structure” AND “well-being”; “polycentric” AND “well-being”; and “high-rise” AND “well-being”. For subtopics where literature on their relationship to well-being was limited, more generic searches were performed, for example “urban” AND “vitality”; “15-minute city” OR “x-minute city”; and “urban” AND “artificial intelligence”. All searches were conducted filtering the year of publication from January 2021 to present (this was February 2026, the month during which the search was conducted).

The selection of studies for review was guided by relevance to the conceptual framework of pathways linking the built environment and SWB (presented in Section 3) and by the paper's aim of synthesizing recent research advances. Priority was given to studies presenting thematic, empirical, or methodological contributions relevant to urban planning and policy. These contributions are referred to here as “advances” and included emerging research themes, new conceptual perspectives, substantial new empirical evidence within established themes, and new evidence based on methodological developments such as longitudinal approaches, natural experiments, and literature reviews. The review focuses primarily on studies examining built environment characteristics – mainly urban form and land use, transport systems, architecture and public space, housing, and digital and ICT infrastructure – in relation to SWB and its components life satisfaction, emotional well-being, eudaimonia, relevant concepts such as quality of life, and life domains that represent determinants of subjective well-being including health, work, mobility, leisure, and social relationships. Moreover, the review focuses on studies of emerging urban themes with promising

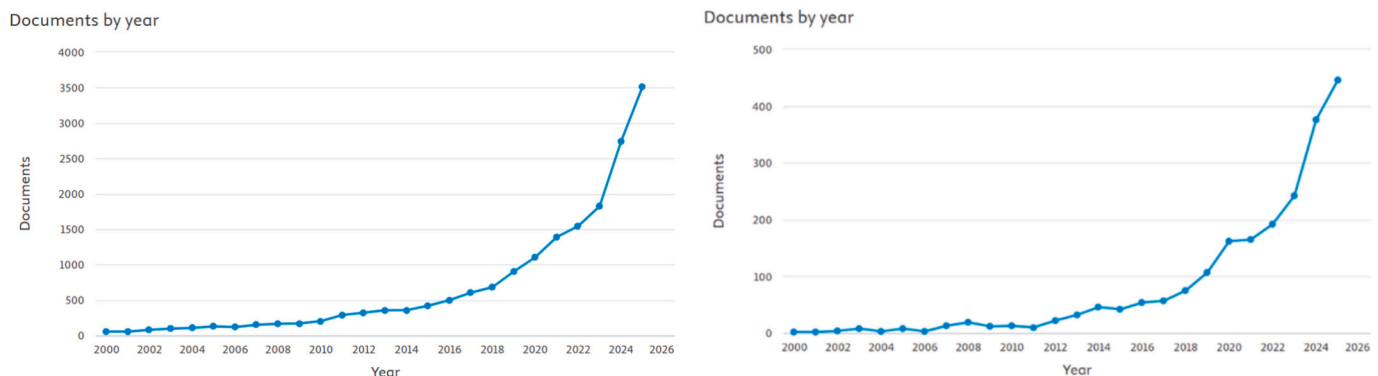


Fig. 1. Left: “urban” and “well-being” search outputs. Right: “built environment” and “well-being” search outputs. Notes: Period 2000–2025. Search in title, abstract, and keywords. Source: Scopus.

potential for future research in relation to SWB (e.g. 15-minute city; urban vitality; urban artificial intelligence). Studies were generally excluded when they lacked substantive relevance to built environment-SWB pathways or when they focused on outcomes without implications for SWB.

The review is intentionally selective rather than exhaustive, with the aim of synthesizing major recent conceptual, empirical, and methodological advances across diverse themes within the built environment and SWB literature. Naturally, due to the wide scope of the review, there are research advances and relevant studies not reviewed in the paper. The reviewed studies span across a wide range of geographical contexts, including substantial recent contributions from China, Europe, North America, Australia, and Japan. While the review does not aim for geographical representativeness, it aims to capture influential developments and emerging perspectives across different settings.

3. Conceptual model and overview of advances

The review relies on the conceptual model of “pathways linking the built environment to SWB” by Mouratidis (2021b). The model has been updated to reflect recent trends in literature (Fig. 2). The updated model in Fig. 2 includes only minor revisions compared to Mouratidis (2021b) and is primarily used here as a conceptual framework for organizing and interpreting recent research advances. The revisions aim to improve conceptual clarity and inclusiveness rather than introduce a fundamentally new theoretical model. Specifically, “land use” has been expanded to “urban form and land use”; “urban design” has been replaced by “architecture and public space” to reflect the relevance of both architecture and public space for urban well-being; “digital and ICT infrastructure” and “basic urban infrastructure” have been added to reflect the relevance of digital systems and essential infrastructure for well-being; and the concept of “travel” has been replaced by the broader concept of “mobility” that includes both the potential to travel and actualized trips.

An explanation of the model (Fig. 2) is presented here. The built environment is the physical environment made or shaped by humans. It includes: urban form and land use; transport systems; architecture and public space; housing; digital and information and communication technology (ICT) infrastructure; and basic urban infrastructure including water, waste, and electricity (Mouratidis, 2024a). SWB is a concept used to refer to a subjective and holistic evaluation of quality of

life (Diener, 2009; Veenhoven, 2012) and is typically measured as life satisfaction and emotional well-being (also called affect), and, additionally but more rarely, eudaimonia (OECD, 2013). Here the three dimensions of SWB are used. Life domains represent key aspects of quality of life that are influenced by the built environment and in turn contribute to well-being. These life domains are mobility, leisure, work, social relationships, residential well-being, emotional experience, and health. The impacts of the built environment on SWB through these life domains constitute the pathways linking the two concepts (Mouratidis, 2021b). Life domains are interlinked and built environment characteristics may influence several life domains simultaneously. Understanding the pathways from the built environment to SWB provides important insights for urban planning and policy relevant to social sustainability and livable cities and communities (Mouratidis, 2018b).

The pathways model adopted here (Fig. 2) is grounded in the urban planning and well-being tradition, drawing on previous theoretical frameworks connecting the urban environment to human well-being through life domains that include perceptions, behaviors, and experiences (Campbell, Converse, & Rodgers, 1976; Marans, 2003; Marans & Rodgers, 1975; Marans & Stimson, 2011; Mouratidis, 2018b). The model has some limitations that should be noted. First, it represents pathways in a simplified, unidirectional form and does not portray bidirectionality. SWB and life domains also influence residential location choices, mode choices, and housing decisions. SWB also influences life domains. So, the relationships in Fig. 2 could also been displayed as bidirectional. Second, as the aim is to show how life domains mediate the relationship between the built environment and SWB, the components of the model are simplified in Fig. 2 and do not show detailed built environment characteristics or perceptions, behaviors, and experiences linked to life domains. Third, the model does not distinguish between short-term and long-term effects of the built environment on life domains and SWB, which may differ substantially. These limitations should be considered when using the model in theoretical discussions or empirical analyses.

Recent advances in research on the built environment and SWB are summarized in Fig. 3. The advances also include subtopics: concrete developments in built environment and SWB research. The categorization of advances presented in Fig. 3 was developed iteratively during the review process. First, recurring and emerging subtopics in built environment-SWB research since 2021 were identified through the literature review as explained in Section 2. These subtopics were then grouped thematically into broader categories of advances based on

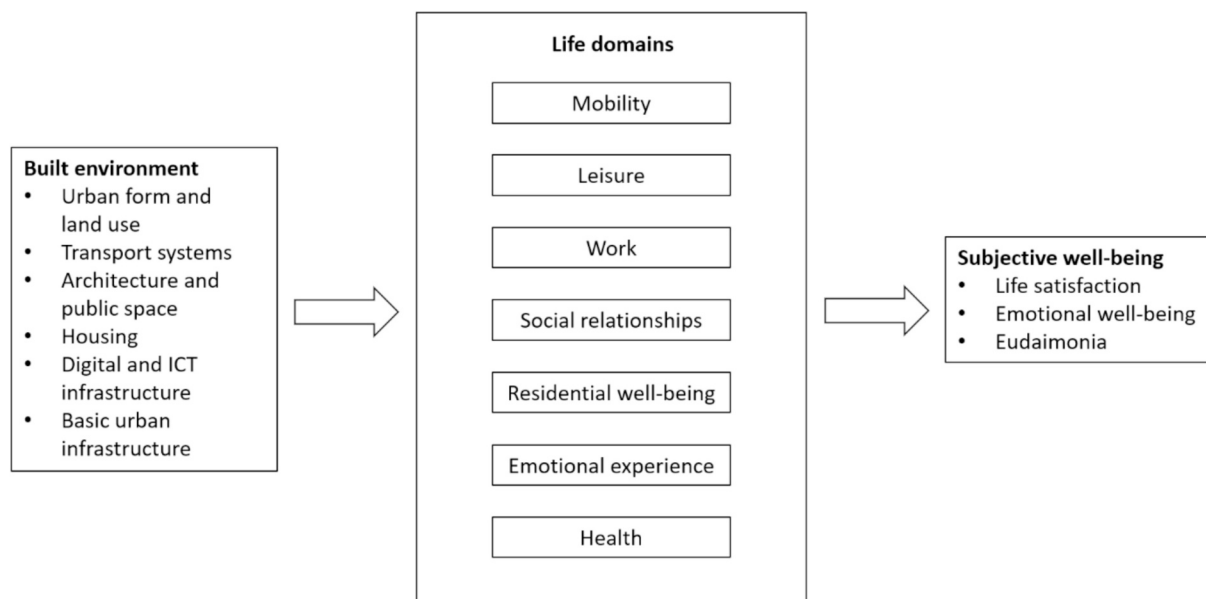


Fig. 2. Model of the pathways between the built environment and subjective well-being (SWB). Updated model based on Mouratidis (2021b).

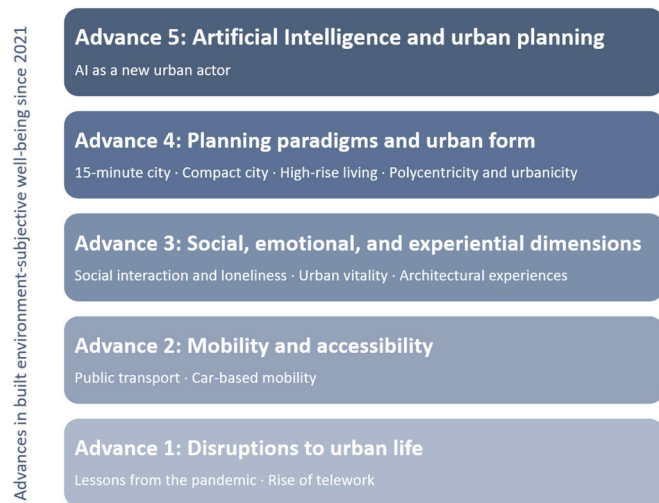


Fig. 3. Overview of advances in research on the built environment and subjective well-being (SWB) since 2021.

shared themes and their contribution to understanding pathways between the built environment and SWB. The resulting categories therefore represent broad clusters of conceptual, empirical, and methodological developments in literature. As explained above, the list of advances and subtopics is not exhaustive. Advance 1 refers to disruptions of urban life and their role in SWB. Advance 2 refers to advances in mobility and accessibility with regard to SWB. Advance 3 refers to advances in social, emotional, and experiential dimensions of the built environment. Advance 4 refers to advances in how planning paradigms and urban form contribute to SWB. Advance 5 refers to generative artificial intelligence as an emergent urban actor that reshapes urban planning and potentially SWB. Among the advances reviewed, artificial intelligence as an urban actor (Advance 5) has a particular relation to the conceptual model. As elaborated below in [Section 8](#), it functions as a modifier of the built environment by shaping urban governance, planning processes, policies, and design outputs and as a contextual factor that modifies most, if not all, aspects of life, including life domains such as work, mobility, social relationships, and health. The five advances are reviewed in [Sections 4–8](#) that follow and are discussed in the context of the pathways as per the conceptual model of [Fig. 2](#).

4. Advance 1: disruptions to urban life and subjective well-being

4.1. Urban planning lessons from the pandemic

The Coronavirus disease pandemic (COVID-19) disrupted daily life across the globe and showcased the importance of urban planning and policy by revealing how housing conditions, access to space, transport systems, and digital infrastructure shape SWB. At the same time, the pandemic created conditions for natural experiments studying the built environment and well-being under pre-, mid-, and post-pandemic conditions. Built environment research focused on indoor environments, green spaces, transport, and facilities and services as indicated by relevant literature reviews ([Mouratidis, 2021a](#); [Parzniewski et al., 2025](#)).

Housing conditions: The importance of living in a dwelling of sufficient size and quality was highlighted during the pandemic ([Mouratidis, 2021a](#)). Overcrowded and poorly ventilated dwellings are conducive to more infections, whereas larger dwelling size facilitates activities performed at home including social activities, leisure activities, and remote activities such as telework. In these ways, dwelling size and quality play a role not only in affective experience but also in other key life domains such as health, social relationships, work, and leisure. Empirical evidence suggests that larger dwelling size was linked with higher life

satisfaction and happiness ([Mouratidis, 2022](#)) as well as better overall health and lower anxiety ([Mouratidis & Yiannakou, 2022a](#)) during the pandemic. Housing conditions did not influence well-being in isolation but interacted with private and public outdoor space that also contributed to well-being as follows.

Green space and public space: The pandemic demonstrated that access to green and public spaces in proximity plays an important role in enabling participation in physical, social, and cultural activities and providing positive affective experiences, particularly during periods where mobility and indoor activities are restricted or avoided ([Mouratidis, 2021a](#); [Nigg, Petersen, & MacIntyre, 2023](#); [Noszczyk, Gorzelany, Kukulka-Kozielec, & Hernik, 2022](#)). Besides public spaces, private outdoor spaces, such as gardens and balconies, also supported such activities and experiences under safe settings. Moreover, besides access, views of green or blue environments from the home offer stress-reducing and restorative benefits, particularly during prolonged stays at home ([Ugolini, Massetti, Pearlmutter, & Sanesi, 2021](#)). Research findings from the pandemic show that access to and contact with blue-green spaces (private or public) were associated with less sharp declines in well-being outcomes ([Cheng, Zhang, Wei, & Zhao, 2021](#); [Lehberger, Kleih, & Sparke, 2021](#); [Mouratidis & Yiannakou, 2022a](#); [Noszczyk et al., 2022](#); [Poortinga, Bird, Hallingberg, Phillips, & Williams, 2021](#); [Pouso et al., 2021](#)). Green spaces and public spaces contributed to well-being also by interacting with transport systems as residents relied on transport systems to access these spaces, as explained below.

Transport and land use: Travel-related risks and experiences varied by transport mode and context during the pandemic. The use of public transport was often associated with higher perceived health risks, stress, and longer commutes ([Hu & Chen, 2021](#); [Mouratidis, 2022](#); [Mussone & Changizi, 2023](#)). Conversely, walking, cycling, and driving alone were generally perceived as safer and less stressful and substituted public transport use when possible ([Schaefer, Tuitjer, & Levin-Keitel, 2021](#); [Shakibaei, de Jong, Alpkökin, & Rashidi, 2021](#); [Vallejo-Borda et al., 2022](#)). Access to a car enabled traveling longer distances to participate in activities under a lower infection risk ([Mouratidis, 2021a](#)). Therefore, people who had access to a private car experienced fewer disruptions to their daily mobility compared to those who relied on public transport ([Sunio, Carlo Ugay, Li, Joy Liwanag, & Santos, 2023](#)). While access to a private car mitigated well-being decline during the pandemic, these findings raise questions about how to reconcile short-term well-being benefits of car ownership with long-term sustainability and equity goals (this is further discussed in [Subsection 5.2](#)). Walkable and bikeable environments played an important role in enabling low-risk mobility and physical activity, particularly when car use or public transport use was restricted or not possible. Walk-based accessibility to facilities and services was associated with positive health and well-being outcomes during the pandemic ([Mouratidis, 2022](#); [Mouratidis & Yiannakou, 2022a](#)). This highlighted the significance of proximity-based accessibility which triggered public debates and policy initiatives such as the 15-minute city ([Moreno, 2024](#)), which is discussed in more detail in [Section 7](#). Besides physical mobility enabled by transport systems, another form of mobility became prominent during the pandemic. That was virtual mobility, enabled by information and communications technology (ICT) as explained below.

Information and communications technology (ICT): The pandemic initiated a massive boost to the use of the internet for participating in activities that would otherwise be performed physically within the built environment. It triggered a sharp rise in virtual activities – including telework, online shopping, online meetings, and e-learning – replacing in-person activities ([Adibfar, Gulhare, Srinivasan, & Costin, 2022](#); [Asmussen et al., 2024](#); [Mouratidis & Papagiannakis, 2021](#); [Mouratidis & Peters, 2022](#)). ICT had to grow substantially to enable participation in such activities. Nevertheless, cities and regions without sufficient access to ICT or individuals without the means or the ability to use ICT suffered greater disruptions in daily life with possibly negative well-being outcomes, thereby highlighting social and spatial inequalities

(Maatuk, Elberkawi, Aljawarneh, Rashaideh, & Alharbi, 2021; Mouratidis, 2021a; Young, Soza-Parra, & Circella, 2022). These patterns show that digital and ICT infrastructure has become an integral component of both the built environment and digital space and that digital access and literacy constitute important components of urban well-being. ICT interacted with housing conditions as residents with larger dwellings and sufficient access to ICT were in better position to substitute in-person activity participation than those living in small dwellings or dwellings without proper access to ICT.

Across the domains reviewed above, three cross-cutting patterns are observed. First, the *availability of and the proximity and accessibility to physical space* (dwelling size, private outdoor space, and nearby green and public areas) functioned as a key determinant of well-being when mobility was restricted. Such physical space influenced the degree to which people could participate in activities, engage in social interaction, and experience positive emotional responses. Second, the pandemic forced *alternative pathways to urban well-being*. When public transport became unsafe and recreation or interaction in indoor spaces got restricted, people turned to other options such as shifting to private car use, using local public spaces and green spaces, and performing virtual activities. The availability of these options was itself influenced by the built environment, e.g. land use, transport system, public space, and ICT infrastructure. Third, *access to these alternative pathways was unequally distributed* across socioeconomic groups and urban contexts, meaning that the well-being cost of the pandemic was often experienced disproportionately by disadvantaged populations.

These patterns have direct *implications for urban planning and policy*. The evidence highlights the importance of cities offering proximity-based accessibility as well as diversity in terms of transport systems, public spaces, and green spaces. Such attributes may make urban areas more resilient to disruptive events. Moreover, evidence highlights the importance of addressing urban inequalities in terms of housing affordability, transport, and accessibility to public space and facilities to improve quality of life and life prospects of disadvantaged residents both under normal conditions and major disruptions. Finally, the lessons from the pandemic indicate a tension between sustainability and resilience during such crises since larger dwellings and private car use, both of which are in contrast with sustainability goals, provided benefits that shielded residents from a sharp decline in well-being during lockdowns.

4.2. The rise of telework

Telework has become an emerging theme in urban and well-being research since it is linked with changes in the built environment, travel patterns, and well-being outcomes. It is associated with several benefits connected to SWB such as improved satisfaction with travel, satisfaction with work, better job performance, and lower likelihood of changing jobs (Bloom, Han, & Liang, 2024; Bloom, Liang, Roberts, & Ying, 2014; Versigghel, Hook, De Vos, & Witlox, 2026). In contrast, frequent telework – three or more days per week – has been found to be associated with increased loneliness, indicating that performing work remotely reduces social interaction and might contribute to social isolation (He et al., 2026).

The built environment influences telework, while at the same time telework triggers changes in the built environment. Obviously, physical ICT infrastructure in the built environment is necessary for telework adoption. The importance of housing characteristics is highlighted for those who work remotely, with better housing quality being linked to positive well-being and performance outcomes (Manu & Rysanek, 2025; Mayer & Boston, 2022). At the same time, telework may reduce office space requirements, enabling the reuse of offices for other purposes. (Hensher, Wei, & Beck, 2023). The neighborhood environment also matters. Access to nearby amenities, including shops and cafés, has been linked to more frequent telework (Mouratidis & Peters, 2022; Yang, Zheng, Wang, Guo, & Zhao, 2025). The role of urban form is not yet clear, with some studies finding a positive relationship between

neighborhood density and telework in the year 2020 (Mouratidis & Peters, 2022) and others finding a negative one in the year 2022 (Hook, Borremans, & Witlox, 2024). These conflicting findings might reflect contextual and methodological differences across studies such as differences in the choice of study area (Oslo, Norway versus Flandres, Belgium), period of measurement (2020 versus 2022), and control variables (controlling for residential location or not).

Telework, as well as other remote online activities, enable virtual mobility while they reduce the need to travel to access destinations (Kappler & de Abreu e Silva, 2025; Mouratidis, Peters, & van Wee, 2021; Wang & Mokhtarian, 2024; Wöhner, 2022). Because of that, telework may also enable relocation to suburban or more remote areas (Caldarola & Sorrell, 2024; Motte-Baumvol & Schwanen, 2024; Mouratidis et al., 2021; Versigghel, De Vos, & Witlox, 2025). This is not always the case however (see e.g. Motte-Baumvol & Porcher, 2025) as residential mobility depends on local contextual elements such as urban policies, urban form, housing, as well as socioeconomic and cultural factors (de Abreu e Silva & Colaço, 2025; Mouratidis et al., 2021; Mouratidis & Andersen, 2024).

Several overall patterns are observed from the literature on telework. Telework offers well-being benefits such as reduced time spent on commuting, flexibility, and higher job satisfaction, but enjoying these benefits depends on built environment characteristics including housing quality and ICT infrastructure. At the same time, frequent telework poses risks for well-being including reduced social interaction and increased loneliness. Telework is a new key component of the pathways framework. Specifically, it is a new type of work arrangement within the work life domain, which is dependent on the built environment and interacts with the life domains of mobility, residential well-being, social relationships, and leisure. These interactions have an influence on SWB which depends on how benefits and risks play out in a given context. The bidirectional relationship between telework and the built environment is also important from a conceptual standpoint. The built environment enables or discourages telework adoption but telework in turn alters the built environment through reduced demand for office space, possible suburbanization, and changes in transport systems due to changes in mobility. These feedback loops are not visible in the unidirectional pathway model, which aims to simplify how the built environment influences SWB, but should be taken into account in future research. The rise of telework also raises questions for urban planning and policy. Questions include understanding shifting travel demands and rethinking urban mobility, addressing land use changes and adapting office space, controlling suburbanization and urban sprawl, and ensuring that digital and ICT infrastructure and digital education are provided to support remote workers of all sociodemographic groups. Advances in telework also point to underexplored and emerging pathways for urban well-being research. These would revolve around understanding how telework-driven time savings, changes in the built environment, and changes in experiences and behavior independently and collectively influence SWB.

5. Advance 2: mobility, accessibility, and subjective well-being

5.1. Public transport

In recent years, there have been significant research advances shedding light on the role of public transport in life domains and SWB. New metro developments in China (Lin, Nelson, Beecroft, & Cui, 2023) as well as other contexts have given rise to important findings from studies that carry out natural experiments before and after the development of new metro stations. In addition, recent public transport studies have investigated previously underexplored themes.

Recent evidence shows that public transport systems improve *employment* opportunities and contribute to increase household *income* (Alhassan & Anciaes, 2025; Wang, Yang, & Shen, 2025). In contrast, insufficient public transport is linked to lower accessibility and

suppressed travel in rural (Ryan, Vitrano, & Martens, 2025) and urban areas (Mouratidis, 2026a). Although public transport promotes economic activity, new public transport stations and transit-oriented development may attract residents from more affluent areas and may lead to increased housing prices and *gentrification* (Alhassan & Anciaes, 2025; Kandt, van Dijk, & Todd, 2025; Ostrensky, Porsse, & da Frota, 2022; Wang, Zheng, Wu, & Wang, 2022). These findings contrast previous evidence suggesting that there is no significant effect of new public transport stations on displacement of low-income households (Delmelle & Nilsson, 2020). The contrasting findings might be due to differences in local housing markets, housing policies, and the type and size of transport investment. This indicates that transit-led gentrification is contingent on local circumstances and policies (Baker & Lee, 2019). This phenomenon has important consequences for the well-being outcomes of public transport investments since benefits such as increased accessibility and opportunities are lost for the economically vulnerable residents who experience displacement.

Another key area of recent research on public transport and well-being is that of *social interaction*, *social ties*, and *loneliness*. Longitudinal findings from China indicate that high-speed rail may help strengthen social ties and improve individual health (Wang, Yang, & Shen, 2025). A natural experiment on new metro infrastructure in China found that older adults experienced improvements in social engagement that likely contributed to improved outcomes of *eudaimonic well-being* (Du & Sun, 2024). Another experimental study suggested that new metro stations increased social activity for older women but not for older men, thus possibly helping mitigating social isolation and loneliness of older women but not of older men (Sun, Du, & Schwanen, 2026). Another relevant finding from recent cross-sectional studies in China and Japan is that proximity to public transport and the use of public transport are related to greater social capital (Qin & Fukuda, 2023; Sun, Lin, & Yin, 2022).

Advances have also been made in the relationship between public transport and *health*. A simulation study found health benefits from the introduction of a commuter rail line, and specifically a modest decrease in diabetes incidence which can expand disability-adjusted years and life expectancy (Mazumdar et al., 2026).

Another advance comes from recent studies highlighting the positive role of rail transit in shaping *travel satisfaction*. Proximity to a subway and higher subway density have been shown to influence both commute satisfaction and overall daily travel satisfaction through direct and indirect pathways and thus possibly increasing overall *life satisfaction* (Li, Ma, & Tao, 2026; Yin, Cao, & Huang, 2021). The introduction of a new subway line may lead to improvements in residents' travel satisfaction, as suggested by findings from Wuhan, China (Li, Lu, Wang, & Wu, 2026). Beyond network expansion, both local access to rail stations and broader network connectivity have been found to positively contribute to travel satisfaction for commuting as well as non-commuting trips, as suggested in a study from Beijing, China (Yun & Zhai, 2025).

Advances also include research on autonomous public transport in the form of *autonomous buses*. Recent findings suggest that a possible deployment of autonomous buses in areas currently underserved by public transport services could increase accessibility and therefore improve quality of life particularly for those who do not use private vehicles (Mouratidis & Peters, 2025). Such forms of public transport would however require policy and environmental adjustments as well as considerations of economic viability. Recent research has also pointed to downsides of autonomous buses, mainly possible unemployment as well as challenges regarding feelings of safety due to the absence of a driver or host on board (Ariza-Álvarez, López-Lambas, & Soria-Lara, 2023; Mouratidis & Peters, 2025; Nikitas, Vitel, & Cotet, 2021).

Together, these advances consolidate the pathways between public transport systems, life domains, and SWB shown in the model in Fig. 2. In particular, recent evidence strengthens the pathways from public transport to SWB through the following life domains: mobility

(increased accessibility), work (employment opportunities and increased income), social relationships (increased social interaction and reduced loneliness), emotional experience (greater travel satisfaction), and health (reduced risk of diabetes). These pathways are linked to SWB including life satisfaction as well as emotional and eudaimonic elements. Findings also provide insight into the mobility life domain from an equity perspective by showing that accessibility gains from public transport investment may be accompanied by risk of displacement that can undermine well-being for economically vulnerable residents.

5.2. Car-based mobility

Another set of advances in transport research is that of automobility and SWB. Recent advances include new conceptual models, literature reviews, and empirical studies. A new conceptual model on automobility and well-being distinguishes between societal (collective) and individual pathways and captures both positive and negative impacts (Mouratidis, 2026b).

From a societal well-being perspective, recent literature reviews show that automobility is associated with a range of adverse collective effects, such as traffic injuries and fatalities, congestion, extensive use of public space, limited investment in active modes and public transport, and negative impacts on public health from injuries, pollution, noise, heat, and sedentary travel (Miner et al., 2024; Mouratidis, 2025a). At the same time, there are positive collective effects from car-related economic activity that generates employment and tax revenues (Mouratidis, 2025a).

From an individual well-being perspective, review findings show that people who use private cars often benefit from greater flexibility, comfort, access to destinations, and employment opportunities, while in contrast, those who do not use or do not have access to a car may experience significant mobility constraints and mobility-related exclusion in car-oriented systems (Mouratidis, 2025a). Recent empirical studies support these claims, finding that access to a private car and infrequent car use are associated with positive health and well-being outcomes for the individual user, with these outcomes being pronounced in car-oriented settings (Eldér, Vilhelmson, & Thulin, 2023; Hess, 2022; Li, Guan, & Wang, 2022; Mouratidis, 2025b; Saadaoui, Salon, Jamme, Corcoran, & Hitzeman, 2025). Further empirical studies – and particularly experimental ones – are needed to shed more light on the causal pathways between automobility and individual well-being in a variety of contexts.

The conflicting findings between individual and societal well-being impacts of automobility represent an important contradiction in this literature. This contradiction is difficult to quantify, since individual and collective outcomes are measured differently and operate at different scales and time periods. The individual well-being benefits of car use are context-specific and are measured at specific points in time. On the other hand, societal well-being disbenefits are accumulated over time, influence contextual factors such as land use and transport systems that may in turn negatively influence individual well-being, and have an impact on planetary and human well-being at both present and future. These nuances present a conceptual and empirical advance for the urban and transport fields as they highlight that car-related investments may negatively affect several societal outcomes over time while providing benefits to and being preferred by private car users.

The advances reviewed in Section 5 (public transport and car-based mobility) point to several patterns that are observed when public transport and automobility are considered collectively. Public transport and private car use both contribute to SWB through multiple pathways, but with notable differences that extend to equity and sustainability dimensions. Public transport investments provide well-being benefits to the individual as well as to society as a whole, as long as accessibility by public transport is equitably distributed and displacement is prevented. Private car access offers well-being benefits to the individual user, particularly in car-oriented contexts. However, car use also produces

negative collective externalities in terms of health, equity, and environmental outcomes. The findings suggest that reducing car dependence and investing in public transport are complementary goals, but attention should be paid to their unintended consequences, mobility-related social exclusion and transit-led gentrification, which should be addressed with equity-based policies. These considerations highlight the need for urban and transport planning to evaluate transport investments and policies in terms of their impact on life domains and SWB across different population groups.

6. Advance 3: social, emotional, and experiential dimensions of the built environment

6.1. Built environment, social interaction, and loneliness

Thematic research on the built environment and social well-being has seen important advances in recent years, with new empirical studies and updated systematic reviews. A systematic review on the built environment and loneliness identified several elements related to the *built environment* that may help reduce loneliness including: local amenities, public spaces, nature and green spaces, walkability, safe and comfortable public transport, spacious and high quality dwellings, outdoor spaces attached to housing, housing tenure and affordability, neighborhood safety, absence of antisocial behavior, and digital connectivity (Bower et al., 2023). These characteristics may not independently and in all contexts reduce loneliness. A series of interconnected built environment characteristics adjusted to local contexts are required to reduce social isolation. Another systematic review specifically focusing on green space and loneliness confirms this argument by indicating that the association between green space and a reduced risk of loneliness is in the majority of existing studies not significant (Astell-Burt et al., 2022). This suggests that individual built environment elements may not meaningfully contribute to social well-being, but a composite of built environment elements is needed to achieve meaningful outcomes for significant parts of the population. This finding has important links to the pathways framework. It suggests that the social relationships pathway between the built environment and SWB is activated by interconnected built environment elements that together enable social interaction and reduce social isolation. This highlights the need for urban planning and policy to consider the built environment as a system rather than a set of independent elements. It also highlights the need for research studies to capture composite effects in addition to assessing independent built environment elements.

Recent research on urban social sustainability indicates that a composite of indoor and outdoor *meeting places* (or *third places*) may foster social interaction, promote inclusion, and reduce loneliness. Places such as community centers, youth clubs, open public spaces, green spaces, sports facilities, cafes, restaurants, shopping facilities, and public services function as meeting places for social interaction and tie formation and help combat loneliness and exclusion particularly for more vulnerable groups like migrants, older adults, and younger adults (Borowski & Stathopoulos, 2023; Hofstad et al., 2025; Jing, Dahlberg, Canter, & Plater-Zyberk, 2024; Littman, 2022; Mouratidis et al., 2024; Qi, Mazumdar, & Vasconcelos, 2024; Sugiyama et al., 2022; Zahnow, 2024; Zhuang & Lok, 2023).

Looking at *urban form* from a city or metropolitan scale, however, recent empirical studies find that higher density neighborhoods – which typically provide a higher density of third places as well as diversity of third places – are associated with lower social capital (Sun et al., 2022). This confirms previous studies finding that local social ties are weaker in denser neighborhoods (Mazumdar, Learnihan, Cochrane, & Davey, 2018; Mouratidis & Poortinga, 2020) despite these neighborhoods offering more third places and providing more opportunities for making new acquaintances (Mouratidis, 2018a). These findings suggest that higher neighborhood density may increase exposure to other residents and may offer potential for social interaction, but without necessarily

fostering repeated interactions with neighbors that generate familiarity and strong local social ties. Density may thus influence social relationships in both positive and negative ways which are moderated by personal characteristics and contextual factors. Recent evidence sheds more light on the role of urban form by indicating that besides the role of location and density, it is also housing typologies, length of residence, and neighborhood socio-demographic characteristics that may contribute to local social cohesion and sense of belonging (Maya-Jariego, Santolaya, & Pastor-Alcayde, 2026).

Another area with considerable advances is *mobility* and social interaction/loneliness. Community mobility – the ability to move around one's environment – is found to combat loneliness (DeJohn, Liu, Ma, Widener, & Liu, 2025). Conversely, transport-related barriers are found to increase loneliness and depressive/anxiety symptoms (Choi & Marti, 2025). Perceived walkability is positively linked to place attachment, social interaction, and social capital (Hong, Lee, & Ham, 2026; van den Berg, Liao, Gorissen, van Wesemael, & Arentze, 2022). Objectively measured walkability around the residence was found to promote two types of social interaction: tie maintenance and tie formation (Ishikawa, Kyttä, & Rinne, 2026). Besides mobility in the local neighborhood, recent findings point to the role of transport for mid- and long-range distances in social interaction. Specifically, rapid transit (metro), high-speed rail, and proximity to public transport have been recently linked to increased social activity, stronger social ties, possibly combating social isolation and loneliness (Du & Sun, 2024; Qin & Fukuda, 2023; Sun et al., 2022; Sun et al., 2026; Wang, Yang, & Shen, 2025). Virtual mobility – enabled by the built environment in the form of ICT infrastructure, dwelling characteristics, and places for performing virtual activities – shape social interaction and SWB. Virtual activities such as telework and social media use may decrease social interaction and might contribute to social isolation (He et al., 2026; Purba et al., 2023), while at the same time certain social media platforms may provide opportunities for making new acquaintances and interacting with friends (Hsu, Yueh, & Hsu, 2021; Kenyon, Kinakh, & Harrison, 2023; Miguel, Lutz, Perez-Vega, & Majetić, 2024).

6.2. Urban vitality

Urban vitality is an emerging theme in urban research. The term urban vitality is used to describe vibrant urban spaces or spaces with a high level of street activity (Jacobs, 1961; Maas, 1984; Montgomery, 1998; Mouratidis & Poortinga, 2020). Urban vitality literature has been growing exponentially in recent years. A large body of literature has been recently examining built environment characteristics that promote vitality, consolidating preexisting evidence. The strongest determinants of urban vitality are found to be local amenities and points of interest (POIs), density, and proximity to the city center (Chen, Yu, Shu, Yang, & Wang, 2023; Doan, Ma, Chen, & Zhang, 2025; Gómez-Varo, Delclòs-Alió, & Miralles-Guasch, 2022; Jiang, Huang, Zhou, & Chen, 2024; Li, Li, Jia, Zhou, & Hijazi, 2022; Mouratidis & Delclòs-Alió, 2026; Paköz, Yaratgan, & Şahin, 2022; Zhang et al., 2025). Following the patterns of human activity in cities, the levels of urban vitality differ between weekdays and weekends as well as between daytime and nighttime (Chen et al., 2023). Another large set of urban vitality studies, coming mostly but not exclusively from China, focuses on developing new conceptual frameworks and methodologies to measure vitality (Gómez-Varo, Delclòs-Alió, Miralles-Guasch, & Marquet, 2023; Huang et al., 2026; Jiang, Zhang, Liu, & Huang, 2023; Li, Yabuki, & Fukuda, 2022; Liu, Ge, Ye, Wu, & Bai, 2023; Liu, Gou, & Xiong, 2022; Lyu, Angkawisittpan, Fu, & Sonasang, 2025; Wang et al., 2025).

Despite extensive research on urban vitality in recent years, the concept has not been much studied in relation to SWB. This is surprising since a strong link between urban vitality and SWB would provide justification for all this renewed interest in vitality. While preconditions of urban vitality like density, proximity, and amenities have been studied in relation to perceived qualities of the neighborhood (e.g. place

attachment, sense of belonging, perceived safety) and SWB (Madrid-Solorza, Marquet, Fuentes, & Miralles-Guasch, 2024; Mouratidis, 2018a, 2019a, 2019b, 2020b, 2022, 2024b; Mouratidis & Yiannakou, 2022a, 2022b), empirical studies on urban vitality itself and well-being are scarce. One recent relevant study indicated a link between vitality and livability, finding that urban vitality is strongly associated with neighborhood satisfaction and positive emotional response (Mouratidis & Delclòs-Alió, 2026).

Considering the popularity of the concept of urban vitality, there is an important knowledge gap concerning positive and negative pathways between urban vitality and SWB. Urban vitality and SWB are related through complex and indirect pathways. Urban vitality is an outcome of the built environment interacting with the social environment and resulting in activity in urban spaces. This activity may enhance the emotional aspect of SWB through positive affective experiences in public space, or undermine it through noise, overcrowding, and overstimulation, with these effects differing among groups depending on sociodemographic characteristics, attitudes, and preferences. Vitality may also result in economic activity which may have positive implications for the work life domain and in turn life satisfaction. Gaining an in-depth understanding of these pathways represents one of the most promising and underexplored research agendas in urban well-being research.

6.3. Architecture, preferences, and emotional experiences

Recent research suggests that the design of building façades contributes to walking experiences and influences emotional responses in public spaces (Brielmann, Buras, Salingeros, & Taylor, 2022; Brunner et al., 2025). Such emotional responses are important because they have been linked to life satisfaction and happiness (Mouratidis, 2020a, 2024b). A growing body of empirical research has recently examined preferences, visual engagement, and emotional responses in relation to architecture, and particularly building façades.

Preference studies show that traditional building exteriors – characterized by attributes including surface complexity, ornamentation, symmetry, and warmer color palettes – are generally viewed as more attractive than modernist building exteriors (Loodin & Thufvesson, 2022), confirming earlier findings where contemporary architectural styles were less preferred than traditional ones (Herzog & Gale, 1996; Herzog & Shier, 2000; Mouratidis & Hassan, 2020; Stamps, 1999). Recent evidence also suggests that the use of traditional architectural styles could modestly improve public acceptance of high-density urban development (Navarrete-Hernandez, Mace, Karlsson, Holman, & Zorloni, 2021). New research additionally examines the role of color in architectural preferences and finds that softer pastel tones are generally favored over the gray colors commonly found in modernist and contemporary buildings (Strzałkowski & Lewicka, 2024). Moreover, historical and well-maintained places were found to score higher on preference and life (as opposed to lifelessness) than modern or poorly maintained places (Lewicka, Lubiński, Strzałkowski, & Iwańczak, 2026), suggesting that design as well as upkeep matter for perception and preference.

Research on *visual attention* further shows that structures with traditional design characteristics tend to attract and sustain visual engagement to a larger extent than modern styles (Ro & Huffman, 2025), confirm earlier eye-tracking studies that observed longer and more frequent fixations on traditional façades than on modernist or postmodern ones (Hollander, Sussman, Purdy Levering, & Foster-Karim, 2020; Salingeros & Sussman, 2020).

In addition, studies of *emotional experiences* indicate that traditional architecture is linked to perceptions of better aesthetic quality (Sherborne & Malthouse, 2026) and to lower stress and mental fatigue among people using public spaces (Brielmann et al., 2022). These findings provide further evidence on the possible link between traditional architecture and emotional well-being found in earlier studies

(Mouratidis & Hassan, 2020).

These findings extend the pathways framework by strengthening the emotional experience pathway linking the built environment to SWB. Architecture and public space design do not only have a functional role such as providing shelter, walkability, and accessibility, but also influence daily emotional experiences within the built environment, which may in turn contribute to SWB. This suggests that decisions on architectural and public space design have emotional well-being implications that should be more explicitly reflected in public debate, participatory processes, planning regulations, and architectural education and practice.

The three subtopics reviewed in Section 6 (social interaction and loneliness, urban vitality, and experience of architecture) share a common trait: experiences within the built environment and how these influence SWB through diverse life domains such as social relationships and emotional responses. Third places, vibrant streets, aesthetically pleasant buildings, and inviting public spaces contribute to well-being not only through outcomes such as accessibility but also by influencing everyday experiences, interactions, and emotional reactions within the built environment. For planning practice, this means that the experiential and aesthetic qualities of meeting places, buildings, streets, and other public spaces deserve attention in plans, regulations, designs, and participatory processes. From an equity perspective, it is important to consider how these qualities are experienced across different neighborhoods and population groups.

7. Advance 4: planning paradigms, urban form, and subjective well-being

7.1. The 15-minute city

The concept of the “15-minute city” which promotes planning cities based on short travel times to everyday destinations on foot or by bike (Moreno, 2024), represents one of the most popular themes in recent urban research. The concept builds on long-established principles in urban theory, practice, and empirical research (Mouratidis, 2024c), but has nevertheless gained popularity since 2021 (Allam, Bibri, Chabaud, & Moreno, 2022; Khavarian-Garmsir, Sharifi, & Sadeghi, 2023; Marquet, Mojica, Fernández-Núñez, & Maciejewska, 2024; Papadopoulos, Sdoukopoulos, & Politis, 2023). It has then begun to shape urban policy agendas as well as research funding priorities in the urban and transport fields. The concept has also evolved into the broader term “x-minute city”, which includes other time thresholds for different activities and contexts (Knap, Ulak, Geurs, Mulders, & van der Drift, 2023; Logan et al., 2022; Olsen, Thornton, Tregonning, & Mitchell, 2022; Pollack et al., 2024; Thornton et al., 2022).

As the 15-minute city has been gaining momentum, it has been critically examined from a range of theoretical perspectives (Caprotti, Duarte, & Joss, 2024; Marquet, Anguelovski, Nello-Deakin, & Honey-Rosés, 2024; Mouratidis, 2024c; Pozoukidou & Chatziyiannaki, 2021) and research trends have been recently synthesized by literature review studies (Abdi & Vanoutrive, 2026; Arias-Molinares et al., 2025; Carvalho, Farber, Manaugh, & El-Geneidy, 2025; Mahmoudpour & Shirazi, 2026; Omwamba, Rotaris, & Longo, 2025; Pirmoradi, Garshasbi, & Solgi, 2025; Sepehri & Sharifi, 2025). Methodological work has also advanced the 15-minute city as several studies have been proposing indicators and measurement approaches for assessing x-minute accessibility (Knap et al., 2023; Logan et al., 2022; Mouratidis, 2026a; Olivari, Cipriano, Napolitano, & Giovannini, 2023; Poorthuis & Zook, 2023; Sdoukopoulos, Papadopoulos, Verani, & Politis, 2024; Zhang, Zhen, Kong, Lobsang, & Zou, 2023). In parallel, a growing body of empirical research has carried out accessibility analyses by applying 15-minute-city-inspired frameworks across diverse urban contexts (Akrami, Sliwa, & Rynning, 2024; Bartzokas-Tsiompras & Bakogiannis, 2023; Birkenfeld, Victoriano-Habit, Alousi-Jones, Soliz, & El-Geneidy, 2023; Bulloch & Olsen, 2024; Calafiore, Dunning, Nurse, & Singleton,

2022; Contardo Ayala et al., 2022; Elldér, 2024; Guzman, Oviedo, & Cantillo-Garcia, 2024; Knap et al., 2023; Liu, Kwan, & Wang, 2024; Murgante, Patimisco, & Annunziata, 2024; Staricco, 2022).

Although the 15-minute city has been promoted as a solution to improve quality of life in cities, besides reducing environmental impacts, empirical research on the 15-minute and well-being is almost non-existent. Identifying this gap in knowledge is itself a contribution of the review and an important part of a research agenda on urban well-being. Research points to socio-spatial inequalities in x-minute accessibility that could negatively impact health and well-being especially for more vulnerable populations and in more deprived neighborhoods (Calafiore et al., 2022; Dobrowolska & Kopczewska, 2024; Rodrigues, Giannotti, Miranda, & Morgado, 2026). However, studies that directly assess the link between 15-minute or x-minute cities and SWB are lacking. To understand the positive and negative aspects of living in dense, 15-minute urban environments, one could look at prior literature on density, accessibility, and well-being and the broader literature on cities and happiness (see e.g. Mouratidis, 2021b, 2024a, 2024b). At the same time, recent policies applied worldwide offer a good opportunity for natural experiments and longitudinal studies that directly investigate 15-minute or x-minute cities and SWB, and thus potentially producing more robust evidence than the current state of the art.

7.2. Density and the compact city

New research has emerged on a longstanding and key theme for urban planning: the relationship between urban density and SWB. Research studies include updated literature reviews, longitudinal studies, cross-sectional studies, and geospatial analyses. Before reviewing these studies, it should be noted that density is measured differently across studies and is experienced differently across contexts. This complicates direct comparisons of the influence of density. Studies use diverse metrics, including population density, dwelling unit density, and floor area ratio, that capture different dimensions of density and are not always comparable. Building typology also matters for the experience of density. A neighborhood of mid-rise perimeter blocks, commonly found in the urban core of many European cities, may be experienced differently than compactly built high-rise towers in Asia or more sparsely built high-rise buildings surrounded by green space in European suburbs or by parking lots in North American inner cities. Culture also influences the experience of density. What is considered dense for a resident of a North American or Australian suburb may be considered normal or even low density for a Chinese or European urban resident. These differences in measurement and experience of density indicate that findings on density and SWB are context-dependent, and generalizations across contexts should be made with caution.

A review on cities and happiness attempted to synthesize the advantages and disadvantages of density for SWB: “Possible associated benefits are increased access to facilities and services, increased physical activity in terms of walking and cycling, more equitable urban mobility, improved satisfaction with travel, health benefits due to active travel and access to healthcare, and increased social, cultural, and civic activity. Disadvantages of higher density include reduced green space, lack of private gardens, more noise, more local air pollution, less positive affective experience, smaller dwellings, and weaker social ties between neighbors” (Mouratidis, 2024a).

New empirical studies provide further evidence of these advantages and disadvantages. An analysis of pathways between urban compactness and SWB found that compactness may positively influence SWB through increased travel satisfaction, neighborhood satisfaction, and satisfaction with social relationships, while it may negatively influence SWB through lower emotional responses to the built environment (Mouratidis, 2024b). A study of quality of life in Chinese cities suggests that living in compact cities is associated with better self-reported health, with the strength of this association varying across the country's different regions (Sun & Zhou, 2022). GIS-based analyses have shown that density is the

most important driver of walk-based accessibility to amenities and recreational facilities as well as accessibility to other areas through public transport, thus consolidating accessibility as one of the most important benefits of the compact city (Mouratidis, 2026a; Yu et al., 2023). Finally, a study on density and green spaces found that while higher density is associated with lower tree cover, there are still cases of high density and high tree cover, suggesting that with appropriate interventions density and greenery can be combined (McDonald et al., 2023). The diversity of findings on density and SWB indicates that density influences SWB through multiple pathways simultaneously, some promoting and others undermining SWB. The net impact depends on urban design qualities, personal characteristics, preferences, and contextual attributes, among other factors.

Another set of studies has recently focused on densification and its impact on societal outcomes, which are relevant to well-being. Several cities, for example in Europe, are becoming denser due to immigration and compact city policies (Cortinovis, Geneletti, & Haase, 2022). This urban transformation has opened new paths for research on density and compactness (Haarstad, Kjærås, Røe, & Tveiten, 2022). Findings on the relationship between densification and gentrification suggest that while densification may accommodate new low-income residents, it may attract a higher proportion of middle- and high-income residents and at the same time lead to displacement of low-income residents in case their dwellings are demolished (Lutz, Wicki, & Kaufmann, 2024). Moreover, densification has been linked with exclusionary housing markets when housing affordability strategies are not present, with a study suggesting that it is not always clear whether densification generates exclusionary housing by rising housing prices or it is attractive areas that tend to become less affordable and at the same attract real estate investment (Cavicchia, 2023). Finally, a study on long-term residency intentions and quality of life in densified neighborhoods found that residents of newly densified neighborhoods participate less frequently in activities within their community and have weaker local social ties than residents of old compact neighborhoods (Mouratidis & Andersen, 2024). These findings indicate that equity and social cohesion are central concerns in densification processes. The well-being benefits of compact urban form are not equally distributed across neighborhoods, and densification processes may displace low-income residents through demolitions or inflated housing prices. Densification strategies could be therefore combined with measures promoting housing affordability and strengthening local social cohesion.

Longitudinal research on density and well-being before and during the pandemic identified relationships that changed during lockdown periods. Density was found to be negatively associated with life satisfaction in Oslo, Norway and Thessaloniki, Greece during lockdowns (Mouratidis, 2022; Mouratidis & Yiannakou, 2022a). These findings suggest that quality of life in compact cities might decline when mobility is restricted and people need to spend most of their time indoors. While these findings expand previous cross-sectional studies, literature on density and well-being would largely benefit from longitudinal studies and natural experiments that focus on identifying causal effects and causal pathways under normal circumstances.

7.3. High-rise living

The relationship between high-rise living and SWB represents a relatively understudied theme that has recently seen some new research. The relevance of this theme is growing as urbanization progresses and high-rise development expands, particularly in Asia. Similar to density, interpreting evidence on high-rise living and SWB requires considering differences in measurement and context. There is no universally agreed threshold for what defines high-rise. In many Northern European or North American contexts, buildings above seven or eight stories may be perceived as high-rise, while in Asian cities like Hong Kong or Shanghai, this threshold could be considerably higher. Building typology in terms of outdoor space and surrounding urban form also matters for the

experience of high-rise living. Again, compactly built high-rise towers are experienced differently than more sparsely built high-rise buildings surrounded by green space or high-rise buildings surrounded by large parking lots. Culture also matters, influencing the experience of high-rise living, similar to the experience of density. These nuances should be considered when interpreting the findings presented below.

An overview of existing research suggests that high-rise environments have been associated with negative psychological and social outcomes, especially among lower-income residents (Kalantari & Shepley, 2021). However, research findings also indicate that potential negative impacts can be mitigated through design and planning. High-rise buildings may shape mental health and well-being through residents' perceptions and daily experiences. Access to green space, views of greenery or water, and features such as sky gardens can positively contribute to such perceptions and experiences, especially for older adults (Elsadek, Deshun, & Liu, 2024; Ji, Yin, Li, & Zhou, 2023; Yan & Du, 2024). Social factors also contribute to well-being in high-rise environments. Opportunities for community interaction and the presence of shared and accessible communal spaces can strengthen social cohesion and reduce loneliness (Chen, Wang, Jian, & Mo, 2026; Nguyen, van den Berg, Kemperman, & Mohammadi, 2025).

These findings extend the pathways framework by showing that besides urban form attributes such as neighborhood density and land use mix, it is also building typology, and specifically building height, that may contribute to SWB. It does so through life domains including residential well-being, social relationships, and emotional experience. Attributes of high-rise buildings, including access to green views, shared communal spaces, and opportunities for social interaction, may determine whether high-rise living fosters or hampers well-being, particularly for vulnerable groups. Considering the number of people exposed to high-rise living worldwide, the relationship between high-rise living and SWB represents a key research topic in the field of urban well-being. Further research is needed to better understand the potential effects of high-rise buildings and their characteristics on life domains and SWB.

7.4. Polycentricity and urbanicity

SWB has been recently studied in relation to polycentricity (the degree of an area having multiple centers) and urbanicity (the degree of an area being urban) in regional studies. Positive relationships between polycentric regional development and SWB have been found in China and North-Western Europe (Hoogerbrugge, Burger, & Van Oort, 2022; Li, Lyu, Gao, & Lee, 2025). Moreover, residents of more urbanized polycentric regions in North-Western Europe are found to be less satisfied with life than residents of polycentric regions that are less urbanized (Hoogerbrugge et al., 2022). Rural residents tend to be more satisfied with life than urban residents in Europe and the United States (Finnemann, Huth, Borsboom, Epskamp, & van der Maas, 2024; Hoogerbrugge et al., 2022; Le Roy & Ottaviani, 2022; Okulicz-Kozaryn & R. Valente, 2021; Weckroth, Ala-Mantila, Ballas, Ziogas, & Ikonen, 2022), but it has been argued that the size of the regression coefficient is too small for this relationship to be meaningful (Prati, 2024). Moreover, this urban-rural well-being divide does not necessarily hold for urban and rural residents in China who were found to be equally satisfied (Fang et al., 2022). Conversely, a global analysis of a broader set of well-being indicators – not only focusing on life satisfaction – found that urban residents tend to report slightly higher levels of well-being compared to rural residents, while these trends differ across groups and indicators (Counted, Cowden, & Lomas, 2024). Another global study finds that urban versus rural well-being varies depending on the income level of the country, with residents of large metropolises reporting the highest life satisfaction in low-income countries and the lowest life satisfaction in high-income countries (Tassinari & Kleine-Rueschkamp, 2025).

The associations between urbanicity and well-being do not reveal an effect of the built environment on well-being, but may be attributed to

economic development and the aggregation of sociodemographic profiles choosing to live in more urbanized areas (Hoogerbrugge & Burger, 2022; Morrison & Weckroth, 2017; Tassinari & Kleine-Rueschkamp, 2025). In addition, built environment characteristics largely vary within cities and regions and influence well-being through numerous interconnected mechanisms (Mouratidis, 2021b). This diversity is lost in debates and empirical studies on urban versus rural well-being. Ongoing counter-urbanization movements, where people migrate from dense urban areas to rural areas, smaller towns, and villages can provide useful arenas for quasi-experimental studies investigating the effect of settlement size and environmental attributes on life domains and SWB.

Polycentricity has also been recently examined at an urban scale. Panel data from Chinese cities suggest that SWB increases with more polycentric urban development, while this association varies across cities and sociodemographic characteristics (Zhang, Xu, Liu, Huang, & Liu, 2026). The potentially positive impact of urban polycentricity on well-being is attributed to shorter travel times, improved access to public services, and lower housing costs (Zhang et al., 2026). Such mechanisms, and particularly shorter travel times and improved accessibility, provide support to the renewed interest in proximity-based urban planning and decentralization of facilities, advocated by the 15-minute city concept (Moreno, 2024) and scrutinized in Mouratidis (2024c).

The pathways framework can be used in future research to better understand the potential pathways between polycentricity/urbanicity and SWB. Most existing research examines associations between the two without delving into mediating pathways. Possible pathways could be related to the life domains of mobility, social relationships, residential well-being, and emotional experiences and may include aspects such as accessibility to basic facilities and services, community life and social cohesion, and experiences in the built environment. Future research could shed light empirical and theoretical light on such possible pathways.

The four subtopics reviewed in Section 7 (15-minute city, density and the compact city, high-rise living, and polycentricity/urbanicity) share a common trait: urban form and how it influences SWB through life domains. The evidence across all four subtopics indicates that planning interventions may improve living conditions for some residents but not for everyone. While it would be naïve to expect all policies and interventions to be beneficial to everyone, attention should be paid to equity and inclusivity. Policies and interventions should be evaluated with these goals in mind besides increasing overall well-being and limiting environmental externalities. The reviewed literature also highlights important gaps in knowledge. The 15-minute city, polycentricity, densification, and high-rise living are highly relevant in policies and research worldwide; however, empirical research on pathways to SWB through life domains remains insufficient. Longitudinal and quasi-experimental research and research that sheds light on mediating pathways would be particularly useful.

8. Advance 5: artificial intelligence, urban planning, and subjective well-being

Artificial intelligence has started to transform urban planning. Besides techniques like machine learning that have been used for data analysis for quite some time, a new artificially intelligent actor has entered urban planning as well as several other technical and political processes. This new actor is Generative Artificial Intelligence (GenAI). GenAI generates content such as texts, images, videos, voices, music, and code. It is not simply another tool that can facilitate planning processes. By generating content, it can shape governance, policies, designs, and in turn the built environment and the life of citizens. GenAI can thus be considered as a new urban actor that makes a difference in urban planning and development. These claims are embedded in existing literature in the following paragraph. However, it is important to note that research evidence is still developing and most existing studies are

conceptual, exploratory, or limited in scale. Moreover, evidence on the actual effects of GenAI on the built environment and SWB is scarce. Therefore, claims made in this section should therefore be read more as theoretically grounded propositions and as a basis for a research agenda rather than as an overview of mature empirical evidence.

How exactly does GenAI participate in urban planning as an actor? It does so by drafting planning policies and scenarios, generating architectural designs and zoning plans, synthesizing texts, producing visuals, and mediating participatory processes (Du, Ye, Jankowski, Sanchez, & Mai, 2024; Fu, Li, Quan, Yigitcanlar, & Wasserman, 2025; Liu, Yigitcanlar, Mehmood, Corchado, & Fu, 2025; Peng, Lu, Liu, & Zhai, 2024), thus co-producing visions for urban futures, reshaping planning debates, and intervening in decision-making (Cugurullo et al., 2024; Iapaolo & Lynch, 2026). GenAI thus actively shapes both urban governance and the built environment (Caprotti et al., 2024; Cugurullo et al., 2024; Liu et al., 2025; Yigitcanlar et al., 2024). The framing of GenAI as an urban actor draws on actor-network theory (Callon, 1999; Emirbayer & Mische, 1998; Latour, 2005). In urban studies and planning, an actor refers to a human or non-human entity that influences planning processes and affects urban outcomes. As a non-human entity, GenAI can influence processes and outcomes. The degree to which such influence is controlled by human actors is context-specific but also represents a theoretical question that is up to scholarly and philosophical debate. Urban GenAI offers opportunities for more inclusive participation, richer scenarios, effective communication through visuals, and sustainability assessments (Fu et al., 2025; Fu, Wang, & Li, 2024; Moradi & Bilorja, 2025). At the same time, urban GenAI and urban AI in general raise concerns around bias, transparency, and exclusion. GenAI is trained on vast datasets that are characterized by structural inequities and linguistic and cultural biases thus potentially marginalizing underrepresented voices and privileging dominant views (Du et al., 2024; Iapaolo & Lynch, 2026; Sanchez, Brenman, & Ye, 2025; Senadheera et al., 2025).

In urban development, GenAI as an actor may amplify not only biases and exclusion in planning processes but also produce outputs and shape urban futures that perpetuate major global urban challenges. These challenges reduce quality of life, sustain socio-spatial inequalities, and reinforce environmental degradation and climate change. GenAI inherits urban problems of the past and present as input data – for example, polluting urban forms, car dependence, lack of accessibility, mobility injustices, inequities in housing, environmental injustice, and public spaces that do not consider residents' needs.

These concerns connect to established frameworks of social justice within urban and transport studies. From an Urban Justice (Fainstein, 2010) perspective, GenAI risks reproducing existing spatial inequalities by being fed with data representing historically unjust conditions. From a Right to the City (Lefebvre, 1967) perspective, GenAI, by moderating participatory processes and drawing visions for the future, raises questions about whose voice and visions are represented and whose are marginalized. From a Mobility Justice (Sheller, 2018) perspective, AI-generated scenarios may perpetuate car dependence and lack of accessibility for certain population groups, which may deepen mobility inequalities. These concerns should be carefully considered and addressed in the governance and use of GenAI for planning and policy purposes.

Assessing urban GenAI's influence on the built environment and SWB is an important avenue for future research. The pathways through which GenAI may influence SWB operate at two levels. GenAI influences urban governance, planning processes, and outputs and thus shaping the built environment as an urban actor. At the same time, GenAI has a profound effect on all aspects of life, including the life domains of the pathways framework. These two levels are interconnected. GenAI has both a direct impact on life domains as well as an indirect impact through its influence on the built environment. Inequalities and exclusionary effects embedded in GenAI could exacerbate injustices in urban well-being through both mechanisms. All these theoretical propositions represent

an key avenue for future research on urban well-being and human societies in general.

A future research agenda could focus on the following questions: In what ways does GenAI transform planning processes, practices, and outputs? How do these influence the built environment compared to traditional planning? How does GenAI collectively influence SWB through its impacts on the built environment and life domains? How do these impacts vary across urban contexts and population groups? And how could urban governance and planning ensure that GenAI fosters SWB in an equitable way? These questions require interdisciplinary, mixed-methods designs that combine theoretical investigations with multifaceted empirical research in diverse contexts.

9. Discussion and conclusions

This paper has synthesized recent advances in research on the built environment and SWB from the perspective of urban planning and policy. The aim was to identify and present key developments in the field since 2021. Findings indicate that urban well-being research has expanded rapidly in volume as well as in thematic coverage. The conceptual model of pathways linking the built environment, life domains, and SWB (Fig. 2) can be useful for understanding and contextualizing this rapidly developing urban well-being research. The review has presented numerous empirical, methodological, and theoretical developments in Sections 4–8. This section does not repeat these developments but presents three overarching outcomes.

First, recent research provides stronger and more nuanced evidence on several of the *pathways between the built environment and SWB* compared to the state of the art in 2021 (Mouratidis, 2021b). The current review's findings indicate that urban form, housing conditions, transport systems, accessibility to destinations, green and public spaces, third places, architectural and public space design, and digital infrastructure shape SWB through their effects on perceptions, experiences, and behavior related to life domains including mobility, social relationships, work, leisure, emotional experience, residential well-being, and health. These pathways are evident across studies on telework, public transport, automobility, loneliness, urban vitality, architectural experiences, planning paradigms such as the 15-minute city and the compact city, high-rise living, polycentricity, and urbanicity.

Compared to the state of the art in 2021, the advances reviewed here strengthen or enrich the pathways between the built environment and SWB pathways in several ways. The mobility pathway is expanded compared with pre-2021 research by new evidence on telework, on the societal impacts of public transport, on the emergence of autonomous public transport, and on the individual versus collective well-being impacts of automobility. The social relationships pathway is now supported by systematic reviews on urban loneliness and longitudinal and natural experiment evidence, particularly from metro and high-speed rail studies, that was scarce before 2021. The work pathway is also enriched with evidence on telework that substantially grew since 2021 largely due to the effect of the pandemic. The residential well-being pathway is enriched with new evidence on density, densification, urban vitality, polycentricity/urbanicity, and high-rise living as well as pandemic-era insights into the role of dwelling size, private outdoor space, and digital and ICT infrastructure in SWB. The emotional experience pathway is consolidated by new empirical work on architectural preferences and affective responses that show that architecture and public space design can meaningfully contribute to SWB not only through their functional role but also through experiential mechanisms. The health pathway is enriched with new evidence on public transport and physical health and new insights into the role of the built environment in health that emerged during the pandemic period. These are substantive advances that deepen our understanding of how the built environment contributes to well-being.

Second, the literature underscores *inequalities in built environment-SWB* relationships. Attributes including mass transit such as metro and

high-speed rail, access to a private car, housing quality and size, green space availability, proximity to destinations, the presence of third places, and access to digital and ICT infrastructure offer well-being benefits. But these attributes are not universally distributed and their influence matters more for specific population groups than for others (Mensi et al., 2026). The review shows that lacking these attributes may reduce well-being for specific groups. This indicates that urban planning and policy may have uneven well-being effects across population groups and contexts.

These inequalities have direct implications for equity in urban planning and policy. The outcomes of the review suggest that investments and policies related to built environment characteristics such as housing, public transport, public space, green space, and digital infrastructure should be assessed in terms of distributional impacts on life domains and SWB across diverse groups and locations. Planning processes could include participatory approaches which enable the representation of diverse voices in how the built environment is developed and at the same time use equity-based indicators in quantitative analyses for decision making purposes. Such approaches could steer urban development to reduce socio-spatial inequalities in well-being.

Third, recent advances *broaden our conceptualization of the built environment and redefine the forces and actors that shape it*. Digital and ICT infrastructure and emerging transport technologies, such as autonomous vehicles, are now parts of the built environment. At the same time, the rise of telework and digital activities in general, reshape behavior and the built environment. Moreover, the emergence of GenAI influences urban governance, planning, and policy, thus shaping the built environment in numerous ways as a new urban actor. GenAI influences SWB both by shaping the built environment and by having a profound impact on human life including the life domains of the pathways framework.

This expansion of the concept of the built environment has important conceptual implications. Physical and digital space have been increasingly interconnected within well-being research. Work, social interaction, and access to destinations are not only enabled by physical proximity and conventional transport systems but also by virtual mobility which is determined by physical (e.g. cables, cell towers, data centers) and digital (e.g. networks, software, operating systems) features. The profound importance of the virtual world means that the built environment is not only shaped by planners, practitioners, and decision makers, but also by actors related to digital technologies and AI as well as by GenAI as an actor itself, as explained above. These developments should be integrated into existing theory on urban planning and SWB.

The review has focused on research advances since 2021, most directly linked to the relationship between built environment and SWB as conceptualized in the framework of Fig. 2. Important global challenges are strongly related to, or even driving this relationship, but were not explicitly investigated in the review as separate advances to keep the scope more focused and the review material more manageable. Some of these challenges are climate change, ageing of urban populations, and the housing crisis. *Climate change* is a key driver of planning decisions, urban transformation, and urban and transport research, with implications for several of the aspects examined in the review including green space, density, transport systems, and housing. Climate change is also a driver of research on public transport and car-based mobility described in Advance 2 and research on planning paradigms described in Advance 4. The *ageing of urban populations* is transforming housing, transport systems, and public space as well as life domains such as mobility and work, with important implications for SWB. Older adults appear across several sections of this review as a population group whose SWB is affected by the built environment. However, ageing as a demographic driver of urban transformation could have been reviewed as an additional research advance similarly to GenAI. Similarly, the *housing affordability crisis and homelessness* represent important dimensions of housing and the life domain of residential well-being with serious implications for inequality, exclusion, and SWB. Housing affordability and gentrification appear in different sections of the review, but a more

explicit review on the housing crisis and its influence on the built environment-SWB relationship could be carried out in future research.

This review has some methodological limitations. It is a narrative review and does not cover all advances in the rapidly expanding urban well-being literature but identifies a selection of key advances within the context of a conceptual framework on the built environment, life domains, and SWB. The narrative approach enables the identification of broader patterns in urban well-being research that may be lost in more narrowly focused reviews. Future studies could explore other advances in urban well-being research but also carry out systematic reviews of the subtopics identified in the paper. The review uses the pathways framework as a basis for identifying and interpreting literature. This approach may exclude studies with a less apparent fit with this framework, which may introduce bias to the outcomes of the review. Moreover, the iterative search strategy, which was employed since literature is broad and rapidly expanding, is not fully systematic and replicable and relies on inclusive selection and subjective judgment, which may be subject to bias. The review is based only on studies published in English, mainly in international journals, which may underrepresent knowledge coming from specific geographical contexts and particularly from the Global South.

The outcomes of the review have several implications for research. First, there is a need to empirically examine underexplored themes, including urban vitality, the 15-minute city, high-rise living, and the emergence of AI as an urban actor, more explicitly in relation to SWB outcomes and mediating pathways. Second, urban well-being research would benefit from causal research designs that rely on longitudinal and quasi-experimental data. Studies should establish causal pathways between built environment attributes (both independent attributes and composites of attributes), life domains, and SWB outcomes. Third, studies could place greater attention to the distributional aspects of the built environment-SWB relationships across population groups and contexts.

For planning practice, the findings point to several specific implications discussed in Sections 4–8 above. Four overarching implications are presented here. First, to foster urban well-being but also enhance resilience to disruptions, planning and policy should enable proximity-based accessibility to basic facilities and services; invest in walkability, mass transit systems, meeting places, facilities and services, public spaces, and green spaces; and facilitate virtual mobility through ICT and digital infrastructure. Second, transport and housing investments and policies should be assessed in terms of their influence on life domains and SWB across groups and locations. Third, buildings and public spaces should be designed not only with functional and environmental goals in mind but also with the goal of enhancing emotional experience and SWB. Fourth, the governance and use of urban GenAI should integrate well-being and justice dimensions so that GenAI does not perpetuate or enlarge existing social challenges and inequalities.

CRediT authorship contribution statement

Kostas Mouratidis: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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